WORKED SOLUTION

STEP II 1990, Question 14

Mechanics • Elastic Collisions

STEP II 1990, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q14

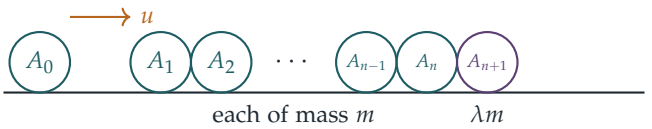
This question has a full guided treatment in *Beyond STEP*, Section 10.1, *Newton's Cradle*: https://beyondstep.keithmcnulty.org/ch10/newtons-cradle.html

The key idea

Without knowing how the impulses pass along the row, we can't follow the collisions one by one. What we can do is test each proposed final motion against three conditions: momentum is conserved, kinetic energy is conserved, and the marbles never pass through one another. The third condition is easy to forget, but it is what decides each part. A marble with a stationary marble directly behind it cannot move backwards, and a marble cannot be moving faster than the marble directly in front of it.

Solution

We model the marbles as particles on a smooth line (the trough), with velocities measured in the direction of A_0's original motion. Before the impact the total momentum is mu and the total kinetic energy is $\frac{1}{2}mu^2$, and both are conserved.



Two facts about any possible final motion follow because the marbles cannot overlap.

- If A_k is at rest and A_{k+1} is directly in front of it, touching it, then A_{k+1} cannot have negative velocity.
- If A_k and A_{k+1} are touching, then A_k's velocity cannot exceed A_{k+1}'s.

Part (i)

Suppose only A_n and A_{n+1} move, with velocities v and w. Dividing through by m,

$$v + \lambda w = u,$$
$$v^2 + \lambda w^2 = u^2.$$

Substituting $v = u - \lambda w$ into the energy equation gives

$$u^2 - 2\lambda uw + \lambda^2 w^2 + \lambda w^2 = u^2 \quad \Longrightarrow \quad \lambda w\big((1+\lambda)w - 2u\big) = 0.$$

The root $w = 0$ gives $v = u$. Then A_n would move forwards into the stationary A_{n+1}, which is impossible. The other root is

$$w = \frac{2u}{1+\lambda}, \qquad v = u - \lambda w = \frac{(1-\lambda)u}{1+\lambda}.$$

If $\lambda < 1$: then $v > 0$, so A_n moves away from the stationary A_{n-1}. Also

$$w - v = \frac{(1+\lambda)u}{1+\lambda} = u > 0,$$

so A_{n+1} moves away from A_n. Both conservation laws hold and nothing overlaps, so this motion is possible, with $A_0, \ldots, A_{n-1}$ all at rest.

If $\lambda > 1$: then $v < 0$, so A_n would have to move backwards. But A_{n-1} is directly behind it, touching it and at rest. So the only motion that satisfies both conservation laws is physically impossible, and no motion of this kind exists when $\lambda > 1$.

Part (ii)

Suppose only A_{n-1}, A_n and A_{n+1} move, with velocities a, b and c. Since $n \geq 3$, the marble A_{n-2} exists, and it is at rest and touching A_{n-1} from behind. So $a \geq 0$. Since the three moving marbles cannot overtake one another, $a \leq b \leq c$. Hence a, b and c are all non-negative.

> **Ask yourself** Both conservation laws give equations in a, b, c. How can we compare them without solving?

We square the momentum equation and subtract the energy equation. Conservation gives $a + b + \lambda c = u$ and $a^2 + b^2 + \lambda c^2 = u^2$, so

$$(a + b + \lambda c)^2 - (a^2 + b^2 + \lambda c^2) = u^2 - u^2 = 0.$$

Expanding the left-hand side,

$$\lambda(\lambda - 1)c^2 + 2ab + 2\lambda c(a + b) = 0.$$

When $\lambda > 1$ and $a, b, c \geq 0$, every term on the left is non-negative, so every term must be zero. In particular $\lambda(\lambda - 1)c^2 = 0$, which gives $c = 0$. Then $0 \leq a \leq b \leq c = 0$, so all three marbles are at rest. That contradicts the momentum equation, since $u > 0$. Hence no such motion exists when $\lambda > 1$.

Part (iii)

The argument in (ii) applies to any set of moving marbles: if every velocity is non-negative and A_{n+1} moves, then the kinetic energy is too small. So when $\lambda > 1$ some marble must move backwards. Every marble except A_0 has a marble directly behind it, so A_0 is the natural candidate for a marble that bounces back.

Ask yourself Which two marbles should we try?

Try A_0 and A_{n+1}, with velocities p and q, and $A_1, \ldots, A_n$ left at rest. These are the same two equations as in (i), with p, q in place of v, w, and the non-trivial solution is

$$p = \frac{(1-\lambda)u}{1+\lambda} = -\frac{(\lambda-1)u}{\lambda+1}, \qquad q = \frac{2u}{1+\lambda}.$$

Momentum and energy are conserved, as in (i). The motion is also physically possible:

- $p < 0$, so A_0 moves backwards away from A_1, and there is nothing behind A_0.
- $q > 0$, so A_{n+1} moves away from the stationary A_n.

So one possible motion is that A_0 rebounds with speed $\dfrac{(\lambda-1)u}{\lambda+1}$, A_{n+1} moves off with speed $\dfrac{2u}{\lambda+1}$, and $A_1, \ldots, A_n$ stay at rest.

This is also what happens if we model the impact as a sequence of separate collisions between neighbouring marbles. Each pair of equal marbles swaps velocities, so the speed u is passed along to A_n. When A_n hits A_{n+1} it rebounds with velocity p, and that backward velocity is then passed back along the row to A_0.

Checks

- When $\lambda = 1$ all the marbles are identical, and both answers become A_{n+1} moving off with speed u while everything else is at rest, which is the familiar "Newton's cradle".
- With $\lambda = 2$ and $u = 3$, part (iii) gives $p = -1$ and $q = 2$. Momentum: $-1 + 2 \times 2 = 3$. Energy: $1 + 2 \times 4 = 9 = 3^2$.
- We stepped through the sequence of pairwise elastic collisions numerically for $n = 3$ and $n = 6$. With $\lambda = 2$ the final velocities were $-\frac{1}{3}u$ and $\frac{2}{3}u$, and with $\lambda = 4$ they were $-0.6u$ and $0.4u$, exactly as predicted. With $\lambda = \frac{1}{2}$ the simulation ends with A_n and A_{n+1} moving at $\frac{1}{3}u$ and $\frac{4}{3}u$, matching (i).

Common slips

- Solving the two conservation equations and stopping there. Each part depends on whether the solution is physically possible, that is, on the signs of the velocities and whether marbles would pass through each other.

- Keeping the root $w = 0$, $v = u$ in (i). That root just describes the motion before the collision.
- Forgetting that $n \geq 3$ is what guarantees there is a stationary A_{n-2} behind A_{n-1} in (ii).

Beyond STEP

For a row of touching marbles, the two conservation laws alone don't fix the outcome. This question is really about which of the many conserved-quantity solutions are physically possible. Choosing between them needs a more detailed model of the contacts, such as a sequence of pairwise impacts or the marbles treated as stiff springs. Modelling the marbles as springs leads to the theory of waves in chains of particles. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers collisions and conservation laws in this spirit.

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